

CENG 201 - Task 4 Report

Topic: Hospital Patient Management – Patient Removal and List Printing

Extend the hospital system to allow removing patients by their ID and printing the complete patient list, while maintaining clean and maintainable code.

Task 4 enhances the hospital system by adding the ability to remove patients by ID and print the full patient list. The PatientList class now includes a removePatient(int id) method to locate and remove a patient and a printList() method to display all current patients. This functionality makes the system more flexible, maintainability, and clarity by keeping patient data organized within the Patient class and allowing dynamic updates without introducing errors. The design follows object-oriented principles, ensuring that operations on the patient list are straightforward and easy to manage.

Implementation Notes:

- PatientList uses an internal list (ArrayList) to store patients.
- Methods ensure safe and dynamic management of patient data.
- Field patientList can be declared final to prevent accidental reassignment.
- All operations are simple and modular for easy testing and debugging.

Time Complexity:

- addPatient(Patient p) $\rightarrow O(1)$
- removePatient(int id) $\rightarrow O(n)$, where n is the number of patients
- searchPatient(String criteria, String value) $\rightarrow O(n)$
- printList() $\rightarrow O(n)$

Conclusion:

Task 4 successfully adds patient removal and list printing functionalities, improving the system's usability while maintaining efficiency and clarity. The operations are linear where necessary, and the code structure supports further extensions without compromising maintainability.